

Photosynthesis: the big picture

Photosynthesis converts light energy into chemical energy stored in glucose.

Plants absorb carbon dioxide from the air and water from the soil.

Using sunlight, plants produce glucose and release oxygen as a byproduct.

The overall equation is $6 \text{ CO}_2 + 6 \text{ H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ O}_2$.

Chloroplasts are the organelles where photosynthesis occurs.

Chlorophyll is a pigment that absorbs mostly red and blue light.

Green light is mostly reflected, giving leaves their green appearance.

Light-dependent reactions

Light-dependent reactions take place in the thylakoid membranes.

Chlorophyll absorbs photons, exciting electrons to higher energy levels.

Water is split to replace electrons, releasing oxygen and hydrogen ions.

An electron transport chain builds a proton gradient across the membrane.

ATP synthase uses this gradient to produce ATP from ADP and phosphate.

NADP⁺ accepts electrons to form NADPH. ATP and NADPH carry energy.

ATP and NADPH power the Calvin cycle. Oxygen is released into the air.

The Calvin cycle and limiting factors

The Calvin cycle occurs in the stroma of the chloroplast.

Rubisco fixes carbon dioxide by attaching it to RuBP.

ATP and NADPH are used to reduce the resulting molecules into G3P.

Some G3P leaves the cycle to form sugars; the rest regenerates RuBP.

The cycle does not directly require light, but uses light reaction products.

Light intensity, carbon dioxide concentration and temperature can limit rate.

Increasing one factor helps only until another factor becomes limiting.

Very high temperatures can reduce the rate by denaturing enzymes.